Fysi Marketplace MVP on Stellar — Product Requirements Document

Nov 6, 2026 · @Laye

Overview

The Fysi MVP is a marketplace for certified eco-friendly products, with all payments settled on the Stellar network. Verified vendors sign up and sell; customers browse and add to cart without an account, then sign up to buy; an admin verifies vendors and monitors activity.

Every vendor gets a custodial Stellar wallet that Fysi manages. Customers choose how to pay: connect their own Stellar wallet, create a custodial Fysi wallet, scan a QR code, or open a payment link.

Item	Detail
Product	Fysi Marketplace MVP (responsive web)
Network	Stellar (testnet for build, mainnet for launch)
Settlement asset	USDC on Stellar; prices shown in naira (NGN)
Roles	Vendor, Customer, Admin
Categories	Clothing, household items, beauty and personal care, eco-friendly gadgets, office supplies, outdoor and travel
Team	Ugochinyere (Project Manager), Chiamaka and Samuel (Product Designers), Steven (Backend Developer)
Source	Fysi case study in Figma

Problem statement

Both sides of the eco-friendly market lack a trusted, dedicated place to meet.

Customers struggle to find a reliable source of curated eco-friendly products. They doubt whether products sold as sustainable really are, and local stores carry little range at high prices.

Vendors struggle to reach buyers who care about sustainability. They rely on social media (65% of surveyed vendors), face high shipping costs and delivery delays, and must manage listings across several platforms.

Payments add friction for both: card and bank-transfer checkouts fail often, settlement to vendors takes days, and cross-border buyers are hard to serve. Building on Stellar gives near-instant, low-fee settlement in stablecoins, with payment by wallet, QR code or link.

MVP scope and success metrics

The MVP proves one loop: a verified vendor lists a product, a customer pays for it on Stellar, and the vendor's wallet is credited within seconds.

In scope

1. Vendor sign-up, business verification, auto-created custodial wallet, product listings, order management.
2. Public storefront: browse, search, product pages, guest cart.
3. Customer sign-up at checkout (cart is kept).
4. Stellar payments: external wallet, custodial customer wallet, QR code, payment link.
5. Admin console: vendor verification, activity monitoring, transaction view.

Out of scope for MVP

- Card or bank-transfer checkout outside Stellar.
- Marketing tools, promotions, discount codes, reviews, wishlist, messaging.
- In-app courier booking (vendors ship themselves and add tracking).
- Advanced analytics, native mobile apps, refunds handled on-chain automatically (manual for MVP).

Success metrics (proposed targets)

Metric	Target
Verified vendors with at least one live product	30 in the first 3 months
Vendor verification decided	within 72 hours
Guest cart to sign-up conversion	⩾ 40%
Checkout to confirmed payment	⩾ 60%
Payment confirmation time	< 10 s after submission
Failed or unmatched payments	< 1%

Metric	Target
Vendor wallet credited correctly	100% of paid orders

Target users and personas

Fysi serves two groups: environmentally conscious customers seeking easy access to eco-certified products, and sustainable vendors who want a green-minded audience. Four personas from the research represent them.

Persona	Type	Profile	Goals	Pain points
Aisha Musa	Vendor	35, eco fashion designer, Lagos	Grow online visibility; widen her range; educate buyers on sustainable fashion	Low visibility, high competition; high shipping costs; managing listings on many platforms
Chinedu Okeke	Vendor	35, eco home and office supplies vendor, Abuja	Reach more customers; cut shipping costs and delays; smooth listing and order management	Shipping costs and delays; low visibility; payment processing and customer service
Grace Ajayi	Customer	32, environmental consultant, MSc Environmental Science, Lagos	Quality eco products for her family; support ethical brands; shrink household footprint	Low local availability; high prices; few trusted online sources
Tunde Adebayo	Customer	28, corporate professional, Abuja	Switch to eco personal care and office supplies without losing quality; support local businesses	Few products in physical stores; limited eco office supplies; unsure of product quality

Behaviours that shape requirements: vendors want marketing support, easy listing tools, certification support, customer management and analytics; customers want clear certification badges, transparent sourcing, and convenience.

Research summary and key insights

The research shows strong demand on both sides, with trust (certification) and logistics as the deciding factors. Data comes from a Microsoft Forms survey of 63 customers (18 questions) and 18 vendors (13 questions), shared via Instagram, Facebook, WhatsApp and vendor email campaigns. Vendor response was lower than hoped, so vendor figures carry more uncertainty.

Vendors

Finding	Figure	Product implication
Sell eco-friendly fashion	60%	Fashion is the anchor category
Rate sustainability verification extremely important	67%	Build certification into onboarding and listings
Rely on social media for sales	65%	Offer a better sales channel plus share tools
Struggle with limited visibility	48%	Featured placement and promotion tools
Concerned about shipping costs	45%	Partner with local couriers
Concerned about delivery delays	35%	Delivery tracking and courier SLAs
Rank marketing support as most important feature	36%	Marketing tools in the vendor portal
Feel neutral about current platforms	47%	Room to win on experience

Customers

Finding	Figure	Product implication
Rate sustainability very or extremely important	84%	Certification visible on every product
Interested in eco clothing and accessories	91–92%	Wide fashion range at launch
Want reusable shopping bags	59%	Household category at launch
Prefer natural soap and shampoo	49%	Personal care category at launch
Want bamboo utensils	44%	Household category at launch

Finding	Figure	Product implication
Prefer reusable office items / recycled notebooks	40% / 32%	Office supplies category
Prefer online marketplaces / websites	40% / 22%	Invest in a robust web platform
Buy eco products occasionally / often / always	43% / 19% / 6%	Large group to convert into regular buyers

Main customer barriers are limited availability and price, so range and competitive pricing matter as much as trust.

Functional requirements

All items below are required for the MVP unless marked "Nice to have".

Vendor

ID	Requirement
V1	Sign up with email and password; verify email
V2	Onboarding wizard: business name, address, phone, category, ID / CAC registration, sustainability certificate upload
V3	Account stays "Pending verification" until admin approves; vendor sees status and rejection reasons
V4	On approval, a custodial Stellar wallet is created automatically with a USDC trustline; no seed phrase shown
V5	Wallet view: available balance, pending (escrow) balance, incoming payments per order, transaction history with Stellar explorer links
V6	Withdraw: send USDC to an external Stellar address, or cash out to a Nigerian bank via an anchor (bank cash-out live from the pilot)
V7	Create, edit, hide and delete products: title, photos, description, price in NGN, stock, category, certification tag

ID	Requirement
V8	Orders list with status: Paid, Processing, Shipped, Delivered, Cancelled; vendor adds tracking number
V9	Email notification on each paid order

Customer

ID	Requirement
C1	Browse home, categories and product pages with no account
C2	Search by keyword and filter by category and price
C3	Product page shows vendor, verified badge, certification, price in NGN with USDC equivalent
C4	Guest cart: add, remove, change quantity; cart saved in the browser and merged into the account at sign-up
C5	Checkout asks guest to sign up or log in (email and password); delivery address required
C6	Choose payment method: connect external wallet, pay with Fysi wallet, scan QR, or get a payment link (see Payments on Stellar)
C7	Order confirmation page and email, with transaction hash
C8	Order history and status tracking; Confirm delivery button (releases vendor payout) and Report a problem within 7 days of delivery
C9	Custodial wallet view: balance, fund by deposit address or QR, transaction history

Admin

ID	Requirement
A1	Vendor verification queue: view documents, approve, reject with reason, suspend

ID	Requirement
A2	Dashboard: new vendors, new customers, orders, payment volume (USDC and NGN), failed payments, per day / week
A3	Activity log: sign-ups, logins, listings created or changed, orders, payments, withdrawals, admin actions
A4	Transaction monitor: every payment with order, payer, vendor wallet, amount, status, explorer link; flag unmatched payments
A5	Product moderation: hide or remove a listing
A6	User management: view, suspend or reactivate vendors and customers
A7	Role-based admin accounts (super admin, admin) as in the Figma user flow

Payments on Stellar

Every order is paid in USDC on Stellar into a Fysi settlement account, matched to the order by a unique memo, then paid out to each vendor's custodial wallet. This supports multi-vendor carts, platform commission and manual refunds.

Wallets

Wallet	Who	Created	Who signs
Vendor custodial wallet	Every approved vendor	Automatically on approval	Fysi backend (vendor approves withdrawals with password + OTP)
Customer custodial wallet	Customers who opt in	On request, from the account page or at checkout	Fysi backend (customer confirms each payment with a PIN)
External wallet	Customers who prefer self-custody	Customer's own (Freighter, LOBSTR, xBull, Albedo, or WalletConnect wallets)	Customer, in their wallet

Wallet	Who	Created	Who signs
Fysi settlement account	Platform	Once, at setup	Fysi backend, multi-signature

Fysi sponsors the XLM reserve and network fees for custodial wallets (sponsored reserves and fee-bump transactions), so users never need to hold XLM.

Payment methods at checkout

Method	How it works
Connect external wallet	Customer connects via a wallet kit; Fysi builds the payment transaction (amount, USDC, memo); customer signs; Fysi submits and confirms
Pay with Fysi wallet	One tap plus PIN; Fysi signs from the customer's custodial wallet; blocked if balance is too low, with a prompt to fund
Scan QR code	Checkout shows a QR encoding a SEP-7 payment request (destination, amount, asset, memo); customer scans with any Stellar mobile wallet
Payment link	Fysi generates a link to a hosted pay page for the order; customer can open it later or send it to someone else to pay; the link carries the same payment request

Rules

1. Prices are set in NGN; at checkout the USDC amount is quoted and locked for 15 minutes. QR codes and links expire with the quote.
2. Each order gets a unique memo; the backend streams the settlement account and marks an order Paid when amount, asset and memo match.
3. Underpaid: order stays pending and the customer is asked to pay the balance. Overpaid or unmatched: flagged for admin and refunded manually.
4. Vendor payout = order total minus commission, held in escrow and released to the vendor wallet when the customer confirms delivery, or automatically 7 days after the order is marked Shipped.
5. Every payment and payout stores the transaction hash and shows a Stellar explorer link.

Funding and cash-out (required for the pilot): customers can fund a custodial wallet, and vendors can cash out to a Nigerian bank, through a SEP-24 anchor. Several NGN on/off-ramps already run on Stellar (for example via Yellow Card, Linkio and the NGNC naira stablecoin); Fysi will integrate one of them via SEP-24 (see Decisions). Sources: Stellar anchors docs, Stellar Community Fund NGN ramp projects.

User flows

Each role has one gate: vendors must be approved, customers must sign up before paying, admins need admin rights.

Vendor: sign up and sell

1. Sign up and verify email.
2. Complete the onboarding wizard and upload documents.
3. Wait for admin review (status shown on dashboard).
    - Rejected: see reason, fix and resubmit.
4. Approved: custodial wallet is created; vendor lists products.
5. Receive order email; wallet shows the payout.
6. Ship, add tracking, mark Shipped then Delivered.
7. Withdraw to an external Stellar address (or bank, when the anchor is live).

Customer: browse, cart, sign up, pay

1. Land on the site; browse or search with no account.
2. Add products to the guest cart.
3. Tap Checkout; sign up or log in. The guest cart merges into the account.
4. Enter delivery address.
5. Pick a payment method:
    - Connect external wallet and sign.
    - Create or use a Fysi wallet and confirm with PIN (fund it first if empty).
    - Scan the QR code with a mobile wallet.
    - Get a payment link to pay later or share.
6. Payment confirmed on Stellar; order confirmation with transaction hash.
7. Track the order in Order history.

Admin: verify and monitor

1. Log in (super admin creates admin accounts).
2. Review pending vendors; approve, reject with reason, or suspend.
3. Watch the dashboard and activity log.

4. Investigate flagged payments; trigger manual refunds.
5. Hide listings or suspend users when needed.

Customer site map (MVP): Home · Catalogue (categories, new in) · Search · Product · Cart · Checkout · Sign up / Log in · Account (orders, wallet, profile) · About · Support (FAQs, contact, privacy, terms).

Non-functional, security and compliance

Holding custodial wallets makes Fysi responsible for user funds, so key security and regulatory clearance come before launch.

Area	Requirement
Key custody	Private keys generated and stored in an MPC wallet-infrastructure provider (see Decisions); never in the app database or logs
Signing	Custodial payments need user confirmation (PIN); withdrawals need password + email/SMS OTP; daily withdrawal limits
Settlement account	Multi-signature; hot balance kept low, sweeps to a cold account
Auth	Hashed passwords, email verification, 2FA for vendors and admins, session timeout
Admin access	Role-based; every admin action written to an audit log
Reliability	Payment watcher recovers from downtime by replaying from the last processed ledger; no order marked Paid twice
Performance	Pages load in under 3 s on 4G; payment confirmed in under 10 s
Accessibility	WCAG 2.1 AA; colour contrast per the Fysi style guide
Privacy	Comply with the Nigeria Data Protection Act 2023
Regulation	Legal review of custodial wallet and digital-asset rules (SEC Nigeria VASP registration, CBN guidance) before mainnet; KYC tiers for custodial wallets

Area	Requirement
Environments	Stellar testnet for development and QA; mainnet only after a security review

Design system: Forest Green (primary), Grey (neutrals), Lime Yellow (accent) colour scales; DM Sans (Bold, Semi Bold, Medium, Regular); one icon set in six styles. New components needed: wallet card, QR payment panel, payment-method picker, transaction row with explorer link, verification status badge.

Release plan, risks and decisions

The MVP ships in four steps, testnet first.

1. **Foundation:** auth, vendor onboarding, admin verification queue, product listings, storefront and guest cart.
2. **Wallets and payments on testnet:** custodial wallet service, external wallet connect, QR and payment links, payment watcher, payouts, admin transaction monitor.
3. **Pilot:** security review, legal sign-off, NGN on/off-ramp via one SEP-24 anchor, mainnet launch with 10–20 founding vendors in Lagos and Abuja.
4. **Next:** marketing tools, reviews, courier integration.

Risks

Risk	Impact	Mitigation
Customers unfamiliar with crypto wallets	Checkout drop-off	Custodial wallet with no seed phrase; plain-language copy ("Pay with Fysi wallet"); QR for mobile users
Customers can't easily get USDC	Few completed orders	Prioritise an NGN on-ramp anchor; fund-wallet guide
Custodial key compromise	Loss of user funds	KMS/HSM or custody provider, limits, multi-sig, cold storage
Regulatory change on digital assets in Nigeria	Launch delay or redesign	Early legal review; keep custody provider swappable
Payment mismatches (wrong memo, wrong amount)	Unpaid orders, support load	Pre-filled transactions and QR/links; flagging and refund tools for admin

Risk	Impact	Mitigation
Low vendor supply at launch	Thin catalogue	Recruit founding vendors directly; waive commission early

Decisions (resolved open questions)

Question	Decision	Why
Settlement asset	USDC only for MVP. No XLM payments; a naira stablecoin can be added later through the anchor	Stable value, supported by every major Stellar wallet and by NGN anchors; XLM price swings would break NGN pricing
Vendor payout timing	Escrow: funds stay in the settlement account and release to the vendor wallet when the customer confirms delivery, or automatically 7 days after the vendor marks the order Shipped. Vendors see it as a "pending" balance	Builds customer trust, which the research ranks first, and makes refunds simple before payout
Commission	5% per order, deducted at payout and shown on each order; 0% for founding vendors in their first 3 months	Low enough to beat social selling costs; free period helps recruit supply
Custody	Use an MPC wallet-infrastructure provider (for example DFNS, already used to sign Stellar payments in Nigerian fintechs), behind Fysi's own wallet service so the provider can be swapped	Faster and safer than building key management in-house; reduces regulatory exposure
NGN on/off-ramp	Integrate one licensed SEP-24 anchor (shortlist: Yellow Card, Linkio, NGNC) and make it live at the pilot, not after	Without an easy way to buy USDC, customers can't check out; vendors need naira cash-out

Question	Decision	Why
KYC	Vendors: full KYB before approval (CAC certificate, director NIN or BVN, address). Customers: external wallet needs email only; custodial wallet Tier 1 (email + phone) capped at ₦300,000 balance, Tier 2 (BVN or NIN via a KYC provider such as Smile ID or Dojah) lifts the cap. Fiat KYC handled by the anchor	Keeps sign-up light while meeting custody rules
Refunds	Manual, by admin, in USDC to the paying address. Valid for non-delivery, damage or wrong item within 7 days of delivery; personal care only if unopened. Fysi pays the network fee (a fraction of a cent)	Escrow means most refunds happen before vendor payout
Accepted certifications	Recognised labels (GOTS, OEKO-TEX, FSC, Fairtrade, Rainforest Alliance, EU Ecolabel, USDA Organic) or SON product certification earn a "Certified" badge. Small and handmade vendors without labels can submit evidence (materials, sourcing, photos) for a separate "Fysi Reviewed" badge	Keeps the trust promise without shutting out small local makers